Mathematics - Undergraduate Level Assessment 10 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. In xyz-space, what are the coordinates of the point on the plane 2 x + y + 3 z = 3 that is closest to the origin?
   (a) (0, 0, 1)
   (b) (3/7, 3/14, 9/14)
   (c) (7/15, 8/15, 1/15)
   (d) (5/6, 1/3, 1/3)
2. Statement 1 | For any two groups G and G', there exists a homomorphism of G into G'. Statement 2 | Every homomorphism is a one-to-one map.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
3. What is the units digit in the standard decimal expansion of the number $7^2$5?
   (a) 1
   (b) 3
   (c) 5
   (d) 7
4. $(1+i)^1$0 =
   (a) 1
   (b) i
   (c) 32
   (d) 32 i
5. The map x -> axa^2 of a group G into itself is a homomorphism if and only if
   (a) G is abelian
   (b) G = {e}
   (c) a^3 = e
   (d) a^2 = a

True / False (10 marks)

Answer True or False

6. True or False: The minimum value of the function f(x) = e^x - cx occurs at x = log(c) when c > 0.

7. True or False: The norm ||x||_p can be expressed as an inner product when p equals 2.

8. True or False: For any two permutations a and b, the order |ab| is always equal to the sum of the orders of a and b.

9. True or False: The set of all n x n non-singular matrices with rational entries is a finite abelian group under multiplication.

10. True or False: In the group (Z,*), the inverse of an element a is simply a-2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Consider polynomials $P(x)$ of degree at most 3, each of whose coefficients is an element of $\{0,1,2,3,4,5,6,7,8,9\}$. How many such polynomials satisfy $P(-1) = -9$? **(A)** 110 **(B)** 143 **(C)** 165 **(D)** 220 **(E)** 286

12. Find $\tan\left(-\frac{3\pi}{4}\right)$.

End of Paper